Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology, Baramati

Autonomous||Approved by AICTE||Govt. of Maharashtra||Affiliated to SPPU, Pune|| NAAC 'A+' Grade & NBA Accredited (Mech., Comp. & Civil)

DEPARTMENT OF MECHANICAL ENGINEERING

LABORATORY MANUAL

SUBJECT: INTRODUCTION TO ROBOTICS & AUTOMATION

[SUBJECT CODE: ME24052]

YEAR: 2025-26

APPROVED BY:

H.o.D. [Mech] **PRINCIPAL**

VALIDITY FROM: ACADEMIC YEAR 2025 – 2026

PREPARED BY: Mr. Shrikant C. Mahadik

Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology, Baramati

DEPARTMENT OF MECHANICAL ENGINEERING

List of Experiments

FACULTY:

SUBJECT: (ME24052) Introduction to Robotics & Automation

YEAR: 2025-26 **CLASS:** SY (AIDS/IT)

1. Study on selection of Actuators and related hardware such as DC motor, Servo motor, Stepper Motor, Motor drivers based on application.

2. Study on selection of Sensors such as IR sensors, Proximity Sensor, Ultrasonic Sensor, White line sensor, Temperature Sensor, Touch sensor, Tilt Sensor, Accelerometer, Gyroscopic Sensor etc. based on given application.

3. Simulation / Demonstration of Cartesian / Cylindrical/ Spherical robot. (RoboAnalyzer / RoboDk Software).

4. Simulation / Demonstration of Articulated / SCARA robot. (RoboAnalyzer / RoboDk Software).

5. Simulation & Demonstration of a simple robotic system using Robo DK / RoboAnalyzer software for material handling systems.

6. Case study on selection & simulation of the appropriate robot system for given manufacturing and non- manufacturing applications.

Mr. Shrikant C. Mahadik
[Lab-In charge]

Vidya Pratishthan's

Kamalnayan Bajaj Institute of Engineering and Technology, Baramati.

Vision

To achieve Academic Excellence through Persistent and Synergic Collaborations amongst all Stakeholders.

Mission

- To ensure holistic development of students as lifelong learners and problem solvers through value-based quality education.
- To motivate faculty to attain the state-of-the-art knowledge and wisdom in their domain and be a facilitator towards cocreation of knowledge.
- To frame and deploy conducive and empowering policies for multifaceted growth of students, faculty and staff to make them contributors towards excellence.
- To partner with industry for mutually beneficial relations to generate employable and deployable workforce.
- To fulfill the aspirations of alumni, parents, society, region and nation at large by generating technically competent and contributing manpower.

Department of Mechanical Engineering,

VPKBIET Baramati

Vision

To inculcate learning culture in students and faculties to meet the current and future technological challenges of industry and society

Mission

- To impart the students with fundamental knowledge of mechanical engineering.
- To provide practical exposure by promoting students for training and internship in related industries.
- Holistic development of the students by inculcating ethical and moral values towards the society and environment.
- To develop association with premier educational institutions, industries and alumni for enhancement of faculty skill.

EXPERIMENT NO. 1

Title:-

Study on selection of Actuators and related hardware such as DC motor, Servo motor, Stepper Motor, Motor drivers based on application.

Objective: -

To understand the selection criteria for different types of actuators, including DC motors, servo motors, and stepper motors, along with their related hardware such as motor drivers. The study aims to explore the operating principles, control mechanisms, and application-specific advantages of each type of motor.

Introduction:-

Actuators are critical components in electromechanical systems, converting electrical energy into mechanical motion. The selection of an appropriate actuator and its corresponding hardware, such as motor drivers, is essential to ensure the optimal functioning of a system. The most common types of actuators are **DC motors**, **servo motors**, and **stepper motors**. Each type of motor has unique characteristics that make it suitable for specific applications. Understanding the properties, control mechanisms, and selection criteria is crucial for designing an efficient system.

In addition to the motors, **motor drivers** play a vital role in controlling the operation of the motors by managing power flow and amplifying control signals. Each motor type requires a specific driver that can handle the motor's voltage and current demands while ensuring smooth operation.

---

1. DC Motor

Characteristics:

- **Operating Principle:** A DC motor converts direct current (DC) electrical energy into rotational mechanical energy. It works on the principle that a current-carrying conductor placed in a magnetic field experiences a force.
- **Speed and Torque:** Speed control in a DC motor is achieved by varying the supply voltage. Torque is directly proportional to the current flowing through the motor. DC motors provide a high starting torque and are suitable for applications requiring continuous or high-speed operation.
- **Types:** There are two primary types of DC motors:

- **Brushed DC Motors:** Simpler construction with brushes that wear out over time.
- **Brushless DC Motors (BLDC):** Higher efficiency and longer lifespan as they lack brushes and commutators.

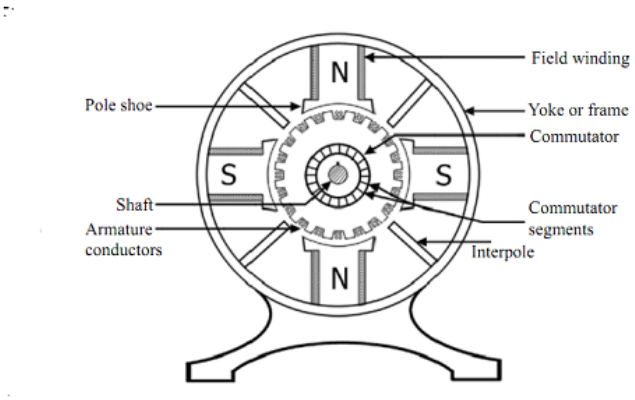


Figure 1: - Construction of DC Motor

Applications:

- **Brushed DC Motors:** Used in electric vehicles, toys, fans, and household appliances.
- **Brushless DC Motors:** Commonly used in drones, computer cooling fans, and electric vehicles due to their high efficiency and precision.

Selection Criteria:

- **Speed and Torque Requirements:** If the application requires continuous high-speed rotation with moderate control over torque, a DC motor is an ideal choice. If precision is needed, other motor types may be more appropriate.
- **Power Supply:** DC motors can run on battery or DC power supplies, making them ideal for applications like electric vehicles and robotics.

---

2. Servo Motor

Characteristics:

- **Operating Principle:** A servo motor is a rotary actuator that provides precise control over angular or linear position, speed, and acceleration. It works based on feedback from a sensor to achieve the desired position. The motor continuously compares the output position with the desired position and adjusts its movement accordingly.

- **Control:** Servo motors use **Pulse Width Modulation (PWM)** to control their angular position. The control signal modulates the motor's output, with a longer pulse corresponding to a higher angular position.
- **Speed and Torque:** Servo motors offer excellent torque at low speeds, making them suitable for precision applications.

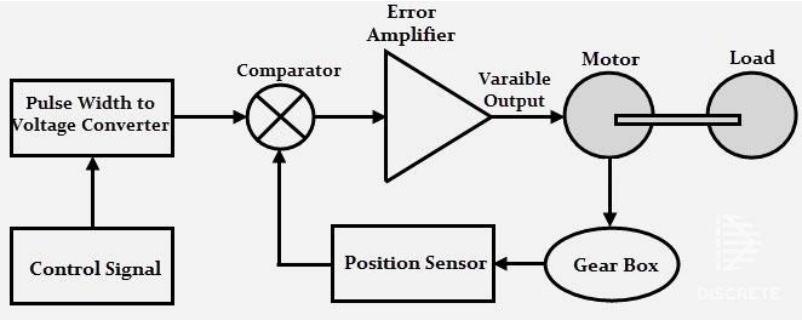


Figure2: - Construction of Servo Motor

Applications:

- **Robotic Arms:** Servo motors provide precise movement and control for each joint in robotic systems.
- **Camera Positioning Systems:** Used to achieve smooth and precise control in pan/tilt mechanisms.
- **Aircraft Controls:** Servo motors are used in flight control surfaces due to their precision and reliability.

Selection Criteria:

- **Precision and Feedback:** Servo motors are chosen for applications requiring precise position control and feedback. For instance, robotic systems that need repeatable and accurate movement benefit from servo motors.
- **Size and Power:** Miniature servo motors are used in low-power applications such as RC planes, while larger servo motors can be used in industrial robotics.

---

3. Stepper Motor

Characteristics:

- **Operating Principle:** A stepper motor divides a full rotation into a large number of steps, with each step corresponding to a specific angle of rotation. Stepper motors operate based on open-loop control, meaning they do not require feedback sensors for basic positioning.
- **Types:** There are several configurations of stepper motors, including:

  - **Unipolar Stepper Motors:** Easier to drive but provide less torque.
  - **Bipolar Stepper Motors:** Offer more torque and efficiency but require more complex driving circuitry.
- **Control:** Stepper motors are controlled by sending electrical pulses to the motor windings. The number of pulses determines the angular movement, and the frequency of the pulses controls the speed.

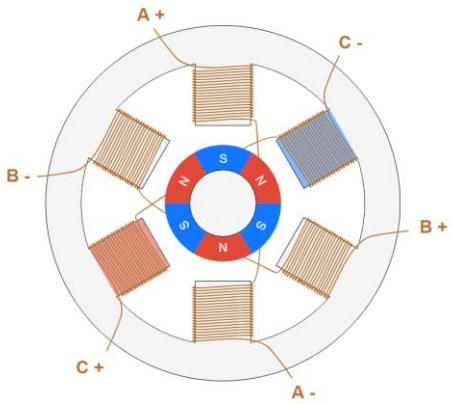


Figure 3: - Construction of Stepper Motor

Applications:

- **3D Printers and CNC Machines:** Stepper motors are ideal for applications requiring precise positioning and repeatability, such as 3D printing, CNC milling, and engraving.
- **Automated Valves:** Stepper motors are used to control the position of valves in industrial automation.
- **Hard Drives:** In older hard drives, stepper motors were used to move the read/write head.

Selection Criteria:

- **Precision and Step Size:** Stepper motors are ideal for applications where discrete, incremental movement is required. The step size determines the resolution of the motor.
- **Holding Torque:** Stepper motors can hold their position when powered, making them suitable for applications where the motor needs to maintain its position even when not in motion.
- **Open-Loop Control:** Unlike servo motors, stepper motors do not require feedback loops, simplifying their control in applications that do not need closed-loop feedback.

---

4. Motor Drivers

Motor drivers are necessary to interface low-power control signals with the high-power motors. The driver amplifies the control signals (typically from a microcontroller) and supplies the necessary current and voltage to the motor.

Types of Motor Drivers:

- **L298N Motor Driver (DC Motors):** Commonly used to drive small DC motors in low-power applications. It provides dual H-bridge configurations for bi-directional control of two motors.
- **A4988 Stepper Driver:** A specialized driver for controlling stepper motors. It allows micro-stepping, improving the smoothness and resolution of stepper motors.
- **ESC (Electronic Speed Controller) for BLDC Motors:** This type of motor driver controls the speed and direction of brushless DC motors. It is widely used in drone and electric vehicle applications.
- **PWM Controllers:** Used to regulate the speed of DC and servo motors by generating pulse-width modulated signals that adjust the effective voltage applied to the motor.

Selection Criteria for Motor Drivers:

- **Motor Type:** The driver must be compatible with the motor (DC, stepper, or servo). For instance, DC motors typically use H-bridge drivers, while stepper motors require specialized drivers like the A4988.
- **Current and Voltage Ratings:** The driver must support the current and voltage levels required by the motor. Selecting an under-rated driver can lead to overheating and failure.
- **Control Interface:** The driver must support the control signals generated by the microcontroller, such as PWM for speed control or step/direction signals for stepper motors.
- **Protection Features:** Advanced motor drivers come with features like over-current, thermal shutdown, and short-circuit protection to enhance durability and safety.
- **Comparison and Application Selection**

Feature	DC Motor	Servo Motor	Stepper Motor
Control	Open-loop (voltage-based)	Closed-loop (feedback)	Open-loop (pulse control)
Precision	Low	High (angular control)	High (step control)
Torque	High	Moderate	Moderate to High
Speed	High	Low to Moderate	Low to Moderate
Holding Torque	No	No	Yes
Application	Fans, Conveyors	Robotic Arms, Cameras	CNC Machines, 3D Printers

Conclusion:-

The selection of actuators and motor drivers is highly dependent on the specific requirements of the application, such as precision, speed, torque, and control. For applications requiring continuous high-speed motion (e.g., fans, conveyors), **DC motors** are the best choice due to their simplicity and high-speed capability. **Servo motors** are ideal for systems that require precise position control, such as robotic arms and camera systems. **Stepper motors** are optimal for applications requiring discrete and repeatable positioning, such as 3D printers and CNC machines.

Motor drivers, equally important, ensure the motor operates correctly by providing the necessary power and control signals. Selecting the appropriate driver ensures that the motor performs optimally without overheating or failure.

By understanding the characteristics and application-specific needs of these actuators and drivers, engineers and designers can choose the correct components to build reliable and efficient systems.

EXPERIMENT NO. 2

Title:-

Study on selection of Sensors such as IR sensors, Proximity Sensor, Ultrasonic Sensor, White line sensor, Temperature Sensor, Touch sensor, Tilt Sensor, Accelerometer, Gyroscopic Sensor etc. based on given application.

Objective: -

The goal is to ensure that appropriate sensors are chosen to meet the specific needs of different systems, such as automation, robotics, environmental monitoring, and user interface controls, to optimize performance, accuracy, and reliability.

Introduction:-

Sensors are critical components in any system that requires environmental interaction, automation, or precise control. A sensor converts physical parameters such as temperature, distance, motion, or pressure into electrical signals that can be read and processed by a microcontroller or processor. The correct selection of sensors ensures the reliability, accuracy, and efficiency of the system. Different types of sensors such as **Infrared (IR) sensors**, **proximity sensors**, **ultrasonic sensors**, **temperature sensors**, **accelerometers**, **gyroscopic sensors**, and more, each have specific uses based on their unique properties and sensing mechanisms.

This study explores the characteristics of various sensors and provides guidelines for their selection based on specific application requirements.

---

1. Infrared (IR) Sensor

Characteristics:

- **Operating Principle:** IR sensors work by emitting infrared radiation and detecting the reflection of the radiation from an object. They typically consist of an IR LED for emission and a photodiode for detection.

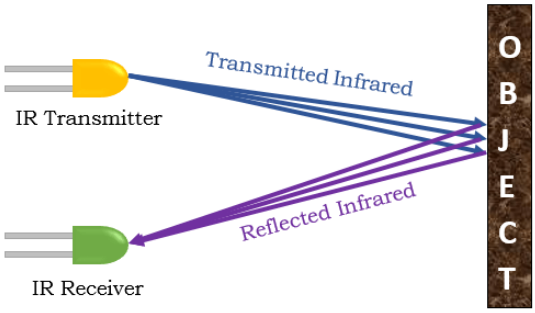


Figure 1: - Working of Infrared (IR) Sensor.

- **Types:**
  - **Active IR Sensors:** Emit and detect infrared radiation (e.g., obstacle detection).
  - **Passive IR Sensors (PIR):** Detect infrared radiation from warm objects, such as human bodies.
- **Range:** Short to moderate, usually within a few meters.
- **Output:** Typically provides a binary output (object detected or not).

Applications:

- **Obstacle Detection in Robotics:** IR sensors are used for detecting obstacles and determining proximity in autonomous robots.
- **Motion Sensing:** PIR sensors are used in security systems to detect human presence.
- **Line Following:** In line-following robots, IR sensors detect the contrast between white and black surfaces.

Selection Criteria:

- **Distance Range:** Select based on the required detection range (e.g., short range for obstacle detection).
- **Environmental Conditions:** IR sensors are sensitive to ambient light, so they may not work well in direct sunlight or reflective environments.

2. Proximity Sensor

Characteristics:

- **Operating Principle:** Proximity sensors detect the presence or absence of objects without physical contact. They use electromagnetic fields or light to sense nearby objects.

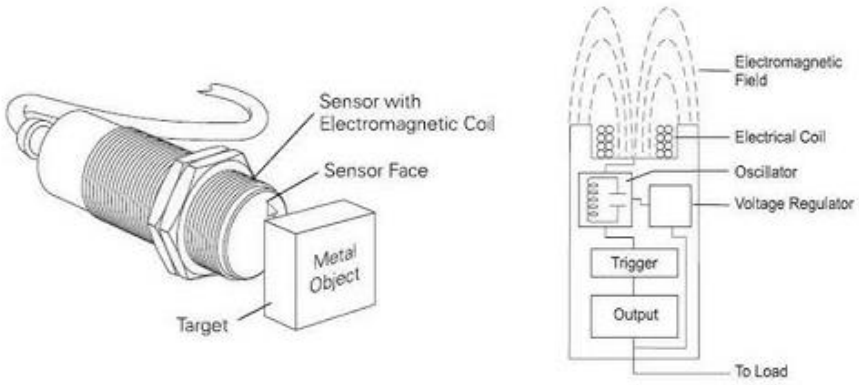


Figure 2: - Working of Infrared (IR) Sensor.

- **Types:**
  - **Inductive Proximity Sensors:** Detect metal objects by generating an electromagnetic field.
  - **Capacitive Proximity Sensors:** Detect both metal and non-metal objects by sensing changes in capacitance.
  - **Optical Proximity Sensors:** Use light to detect objects, often based on IR.
- **Range:** Short range, typically within a few centimeters to a few millimeters.
- **Output:** Binary (object detected or not).

Applications:

- **Automation Systems:** Used in industrial automation for object detection, counting, and positioning.
- **Smartphones:** Proximity sensors are used to turn off the display when the phone is near the ear during a call.

- **Door Automation:** Proximity sensors trigger automatic doors in retail and commercial buildings.

Selection Criteria:

- **Material of Object:** Inductive sensors are ideal for detecting metal objects, while capacitive sensors can detect non-metal objects.
- **Operating Environment:** Choose sensors that can withstand the specific conditions, such as high temperatures, moisture, or dust.

---

3. Ultrasonic Sensor

Characteristics:

- **Operating Principle:** Ultrasonic sensors emit ultrasonic waves (typically 40 kHz) and measure the time it takes for the waves to bounce back after hitting an object. This time is used to calculate the distance between the sensor and the object.
- **Range:** Moderate to long, typically up to a few meters.
- **Accuracy:** Ultrasonic sensors are highly accurate in detecting distance.
- **Output:** Provides analog or digital output proportional to the distance measured.

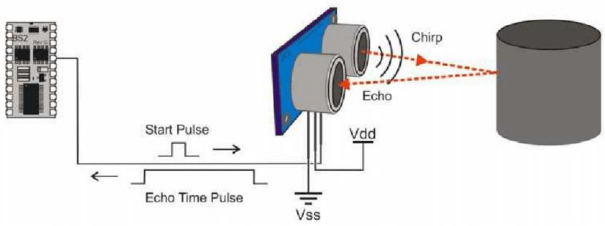


Figure 3: - Working of Ultrasonic Sensor.

Applications:

- **Obstacle Avoidance in Robotics:** Ultrasonic sensors are used in autonomous robots for distance measurement and obstacle avoidance.

- **Distance Measurement:** They are used in applications like parking assistance systems and liquid level measurement.
- **Presence Detection:** Ultrasonic sensors detect the presence of people or objects in a given area.

Selection Criteria:

- **Distance Range:** Ultrasonic sensors are ideal for applications where precise measurement of distance is required over a long range.
- **Surface Material:** The sensor may have difficulty detecting soft or irregular surfaces, which can absorb or scatter the sound waves.

---

4. White Line Sensor

Characteristics:

- **Operating Principle:** White line sensors use infrared reflection to detect light and dark surfaces. The sensor emits infrared light, and a photodiode detects the intensity of the reflected light.
- **Range:** Very short range, typically millimeters to a few centimeters.
- **Output:** Provides a binary or analog output based on the intensity of reflected light.

Applications:

- **Line-Following Robots:** White line sensors are used to follow a pre-defined path marked by a white or black line on a contrasting background.
- **Path Detection:** These sensors help robots navigate along paths in industrial settings or warehouses.

Selection Criteria:

- **Surface Contrast:** White line sensors work best on surfaces with a clear contrast between the line and the background.

- **Precision:** Select a sensor with the required precision for the robot's speed and line complexity.

---

5. Temperature Sensor

Characteristics:

- **Operating Principle:** Temperature sensors measure temperature by converting thermal energy into an electrical signal. Common types include thermistors, thermocouples, and digital temperature sensors.

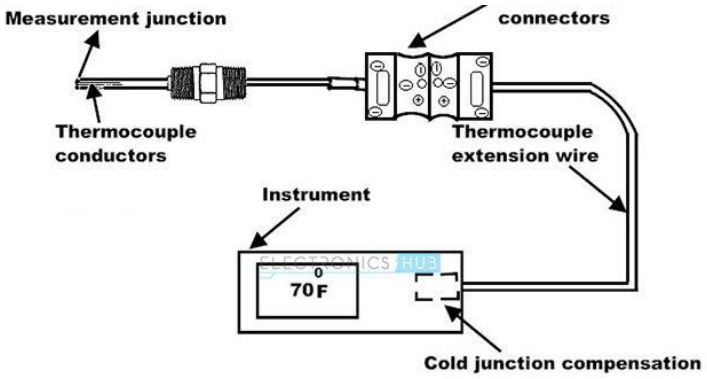


Figure 4: - Working of Temperature Sensor.

- **Types:**
  - **Thermistors:** Resistance varies with temperature, used for simple temperature measurement.
  - **Thermocouples:** Measure a wide range of temperatures using two different metals.
  - **Digital Temperature Sensors:** Provide digital output (e.g., DS18B20).
- **Range:** From very low to high temperatures depending on the sensor type.

Applications:

- **HVAC Systems:** Temperature sensors regulate heating, ventilation, and air conditioning systems.

- **Industrial Processes:** Temperature control in furnaces, refrigeration, and food processing.
- **Consumer Electronics:** Used in smartphones and laptops to monitor internal temperatures.

Selection Criteria:

- **Temperature Range:** Choose a sensor that can measure the required range (e.g., thermocouples for high temperatures).
- **Accuracy and Resolution:** Applications requiring precise control need highly accurate sensors.

---

6. Touch Sensor

Characteristics:

- **Operating Principle:** Touch sensors detect physical contact or proximity to a surface. Capacitive touch sensors are the most common type and work by detecting changes in capacitance when a finger or conductive object touches the surface.
- **Types:** Capacitive, resistive, and piezoelectric.
- **Output:** Provides a digital output (touched/not touched).

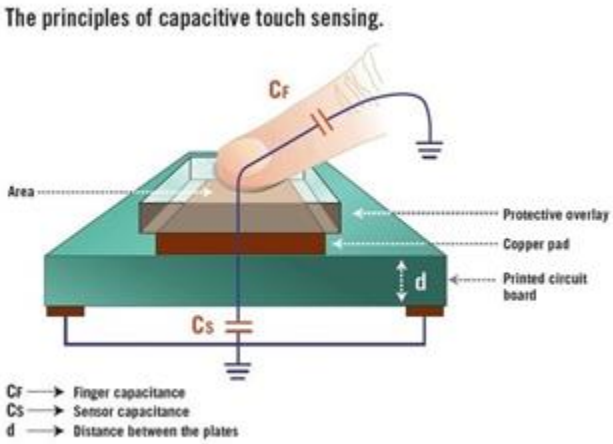


Figure 5: - Working of Touch Sensor.

Applications:

- **Smartphones and Tablets:** Capacitive touch sensors are used in touchscreens.
- **Home Appliances:** Touch sensors are used in microwave ovens, washing machines, and other appliances for user interface control.
- **Wearable Devices:** Used in fitness trackers and smart watches for user interaction.

Selection Criteria:

- **Sensitivity:** Select based on the desired sensitivity for detecting light or firm touches.
- **Environmental Interference:** Capacitive touch sensors can be affected by moisture or water, so choose accordingly for outdoor or wet applications.

---

7. Tilt Sensor

Characteristics:

- **Operating Principle:** Tilt sensors detect the angle of tilt or inclination of an object. They can work using a mechanical mechanism or a liquid-based system that detects gravity's effect.

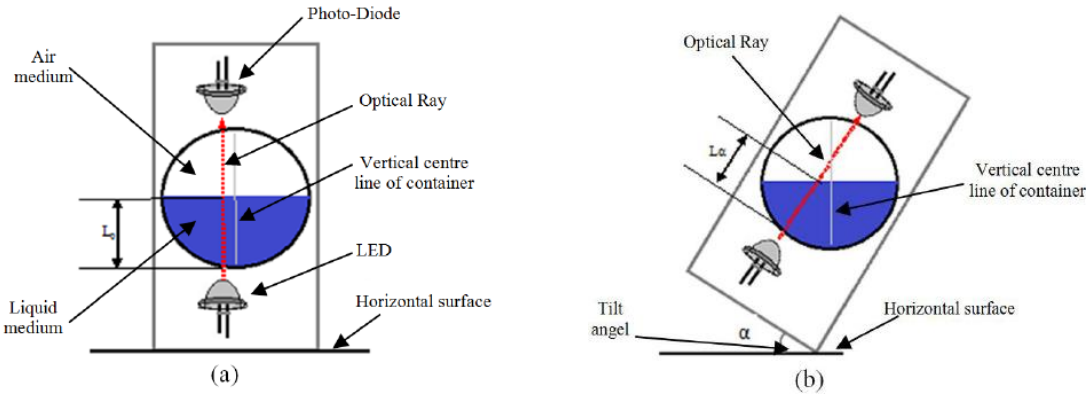


Figure 6: - Working of Tilt Sensor {(a) Initial position (i.e. without inclination) (b) Inclined position by the angel of α}.

- **Output:** Binary (tilted/not tilted) or analog (specific angle of tilt).

Applications:

- **Safety Devices:** Tilt sensors are used in machinery or vehicles to detect if they are at risk of tipping over.
- **Gaming Controllers:** Used in motion-sensing gaming devices.
- **Laptops and Phones:** Tilt sensors detect the orientation of devices for screen rotation or power-saving modes.

Selection Criteria:

- **Angle Range:** Choose based on the required tilt angle for detection.
- **Mounting Orientation:** Some tilt sensors are sensitive to the mounting orientation, so they should be installed appropriately.

---

8. Accelerometer

Characteristics:

- **Operating Principle:** Accelerometers measure the acceleration forces acting on an object, including gravitational force. They are used to measure changes in velocity and orientation.

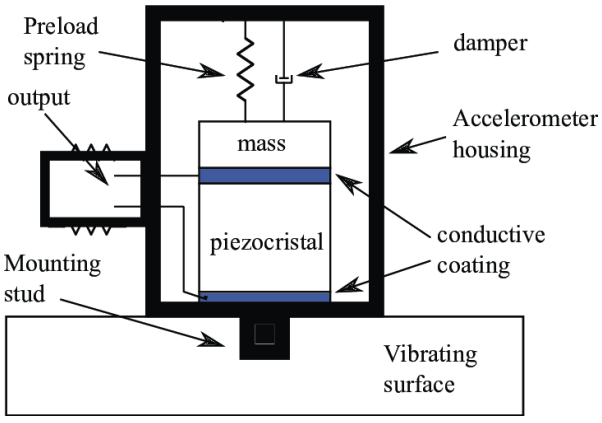


Figure 7: - Working of Accelerometer Sensor.

- **Types:** 1-axis, 2-axis, 3-axis accelerometers.

- **Output:** Provides analog or digital output proportional to the acceleration experienced in each axis.

Applications:

- **Smartphones and Tablets:** Used for detecting screen orientation and motion-based input.
- **Automotive Systems:** Used in airbags and electronic stability control systems to detect rapid deceleration or collision.
- **Fitness Trackers:** Detect body movement and activity levels.

Selection Criteria:

- **Number of Axes:** Choose between 1-axis, 2-axis, or 3-axis sensors depending on the motion complexity required (e.g., 3-axis for tracking motion in all directions).
- **Resolution and Sensitivity:** Higher resolution and sensitivity are needed for detecting small or rapid movements.

---

9. Gyroscopic Sensor

Characteristics:

- **Operating Principle:** Gyroscopic sensors measure angular velocity, providing information on how quickly an object is rotating about its axis.
- **Types:** Mechanical gyroscopes, MEMS gyroscopes (used in smartphones).
- **Output:** Analog or digital output proportional to the angular velocity.

Applications:

- **Inertial Navigation Systems:** Used in drones, airplanes, and spacecraft to maintain orientation and direction.
- **Virtual Reality Systems:** Used in VR headsets to detect head movement.

- **Stabilization Systems:** Used in camera gimbals to stabilize the image by compensating for unintentional motion.

Selection Criteria:

- **Axis of Measurement:** Select based on the required axes of rotation (e.g., 3-axis for full rotational freedom).
- **Drift and Sensitivity:** Higher-end gyroscopes have minimal drift over time and offer greater sensitivity for precise applications.

Conclusion:-

The selection of sensors based on application is critical to the success of any system that relies on environmental interaction, control, or automation. Each type of sensor has unique characteristics, including its operating principle, range, sensitivity, and environmental constraints, which make it suitable for specific applications.

- **Infrared sensors** are ideal for proximity detection and motion sensing but require consideration of ambient light conditions.
- **Proximity sensors** offer contactless detection and are highly reliable in automation systems.
- **Ultrasonic sensors** provide accurate distance measurement and are essential in obstacle avoidance and level detection applications.
- **Temperature sensors** ensure accurate temperature control in various systems, from HVAC to industrial processes.
- **Touch sensors** revolutionize human-device interaction in consumer electronics and industrial controls.
- **Tilt sensors, accelerometers, and gyroscopes** provide vital data for orientation, stability, and motion tracking, particularly in gaming, smartphones, and autonomous systems.

The proper selection of these sensors involves careful analysis of the application's specific needs, including accuracy, range, environmental conditions, and compatibility with other

hardware components. A well-chosen sensor not only enhances the performance of the system but also increases its efficiency, reliability, and overall functionality.

EXPERIMENT NO. 3

Title:-

Simulation of Spherical robot configuration using RoboDK Software.

Objective: -

This study aims to simulate the motion and functionality of **Spherical robots** using RoboDK software. Through this analysis, the focus is on understanding the structural configurations, work envelopes, and industrial applications of these robots, providing insights into their real-world performance for automation tasks.

Introduction:-

Robots are essential in industrial automation, performing tasks that require precision, speed, and consistency. Different types of robots, such as **Cartesian**, **Cylindrical**, and **Spherical** robots, are designed for specific applications based on their motion capabilities, workspace, and the tasks they are intended to perform. Understanding the configuration, motion patterns, and applications of these robots is crucial for selecting the right robot for a particular industrial need.

This study provides an in-depth look at Cartesian, Cylindrical, and Spherical robots, highlighting their structural configurations, modes of motion, work envelopes, and their respective applications in various industries.

---

1. Cartesian Robots

Structure:

- **Linear motion** along three orthogonal axes: X, Y, and Z.
- The structure consists of three prismatic (sliding) joints, providing straight-line motion in three directions.
- The movement is controlled by actuators, typically DC motors or stepper motors with linear guides.

Motion:

- Cartesian robots move in a **rectilinear** manner along the three axes.
- Each axis is controlled independently, allowing the robot to perform precise movements in a cuboidal or rectangular workspace.

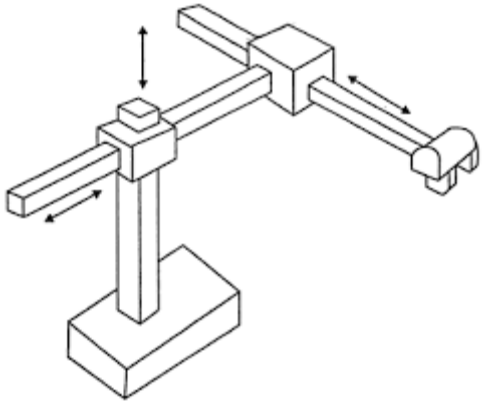


Figure 1: - Cartesian Robot.

Work Envelope:

- **Rectangular or cubic**, depending on the range of each axis.
- The workspace is defined by the travel limits of the X, Y, and Z axes.

Applications:

- **3D Printing**: Cartesian robots are commonly used in 3D printers, where precise linear movement is required to deposit material layer by layer.
- **CNC Machines**: These robots are ideal for Computer Numerical Control (CNC) machines, where accuracy in cutting, drilling, or milling is critical.
- **Pick and Place**: Cartesian robots are used in automation for pick-and-place tasks, where objects need to be moved between fixed positions with high precision.

Advantages:

- High precision due to linear movement.
- Easy to program and control.
- Simple mechanical structure makes them cost-effective for many industrial applications.

Disadvantages:

- Limited flexibility in motion (cannot rotate objects easily).
- Large footprint due to the linear arrangement of axes.

2. Cylindrical Robots

Structure:

- **Rotational and linear movement**: Cylindrical robots have one rotary joint and two prismatic joints.
- The structure consists of:
  - A base that rotates around a vertical axis (θ-axis),
  - A vertical sliding axis (Z-axis) for linear movement,
  - A radial axis (R-axis) for extending or retracting the robot's arm.

Motion:

- The robot can rotate around its base (rotational motion) while extending or retracting its arm (radial movement).
- The Z-axis provides vertical motion, enabling the robot to reach different heights.

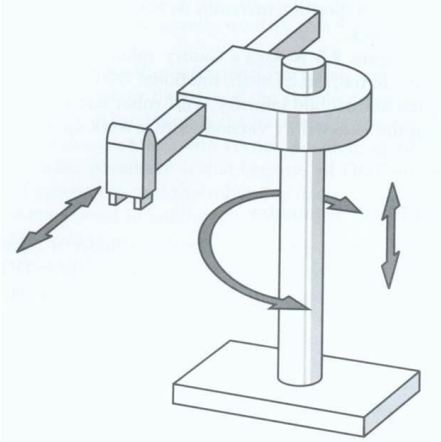


Figure 2: - Cylindrical Robots.

Work Envelope:

- **Cylindrical** workspace, providing motion around a circular area with linear reach in height and radius.
- The workspace is defined by the range of rotation (θ-axis), vertical movement (Z-axis), and radial extension (R-axis).

Applications:

- **Material Handling**: Cylindrical robots are commonly used for moving materials between different heights and radial distances in industrial settings.
- **Welding**: These robots are suitable for spot welding operations, where they can move around the workpiece while adjusting the height and distance.
- **Assembly Lines**: Cylindrical robots are often used in automated assembly processes that require repetitive tasks involving vertical and radial movements.

Advantages:

- Compact design with efficient use of space.
- Suitable for tasks requiring vertical and radial movements.
- Capable of reaching objects at various heights and distances within a cylindrical workspace.

Disadvantages:

- Limited flexibility in terms of angular movement.
- Complex mechanical structure compared to Cartesian robots.

---

3. Spherical Robots

Structure:

- **Rotational and linear movement**: Spherical robots have two rotational joints and one prismatic joint.
- The structure consists of:
    - A rotational joint that allows movement around a vertical axis (azimuth),
    - A second rotational joint that allows movement around a horizontal axis (elevation),
    - A prismatic joint that provides linear movement along the radius (radial extension).

Motion:

- Spherical robots move by rotating around their base (azimuth) and adjusting the angle of their arm (elevation) while extending or retracting their arm radially.
- This motion allows the robot to reach objects in a **spherical** workspace.

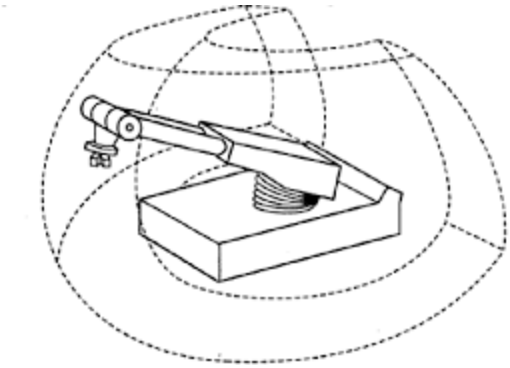


Figure 3: - Spherical Robots.

Work Envelope:

- **Spherical** workspace, allowing the robot to operate within a spherical area around its base.
- The workspace is defined by the rotation angles and the radial distance the robot can reach.

Applications:

- **Robotic Inspection**: Spherical robots are ideal for inspection tasks where they can rotate and extend their arm to reach different angles and surfaces.
- **Robotic Painting**: These robots are used for painting curved or irregular surfaces, where they can move smoothly along the contours of the object.
- **Surveillance Systems**: Spherical robots are used in surveillance and monitoring systems, where they need to cover a wide range of motion in all directions.

Advantages:

- Excellent reach and flexibility in all directions.
- Ideal for tasks requiring wide angular movement and variable distance from the robot base.
- Compact design with good coverage of the workspace.

Disadvantages:

- More complex control and programming due to multi-axis motion.
- Limited to specific applications where spherical movement is required.

Comparison of Cartesian, Cylindrical, and Spherical Robots

Feature	Cartesian Robot	Cylindrical Robot	Spherical Robot
Motion Type	Linear (X, Y, Z)	Rotary and linear (θ, Z, R)	Rotary and linear (Azimuth, Elevation, Radial)
Work Envelope	Rectangular or cubic	Cylindrical	Spherical
Applications	3D printing, CNC, pick-and-place	Material handling, welding	Robotic painting, inspection
Advantages	High precision, simple structure	Efficient use of space, vertical reach	Flexible motion, excellent reach
Disadvantages	Large footprint, limited flexibility	Limited angular movement	Complex control

Simulation of Spherical Robots by using RoboDk

RoboDK is versatile, industry-standard software for robot simulation, programming, and offline programming. It supports a wide variety of robotic models and can be used to simulate both simple and complex industrial tasks.

Features:

- **Robot Programming**: RoboDK generates the necessary code for robots based on the simulated tasks, supporting many industrial robot brands.
- **3D Visualization**: It provides a 3D environment where users can import models, design custom workstations, and simulate robots interacting with real-world objects.
- **Offline Programming**: The software allows for the simulation and testing of robotic programs before actual implementation, reducing downtime and increasing productivity.
- **Multi-Robot Support**: RoboDK can simulate multiple robots working together in the same environment, making it ideal for large-scale industrial operations.

Simulation Using RoboDk:

1. **Robot Setup**: A spherical robot can be modeled in RoboDK by defining the two rotational joints for azimuth and elevation, and one prismatic joint for radial extension. Predefined models for spherical robots are also available in the RoboDK library.
2. **Motion Simulation**: RoboDK allows users to simulate the movement of the spherical robot by specifying radial distances and rotational angles. The software visualizes how the robot arm extends and rotates to reach specific points in its spherical workspace.
3. **Demonstration of Complex Tasks**: Tasks such as painting or surface inspection can be simulated, where the robot must navigate around a 3D object while maintaining specific distances from the surface.

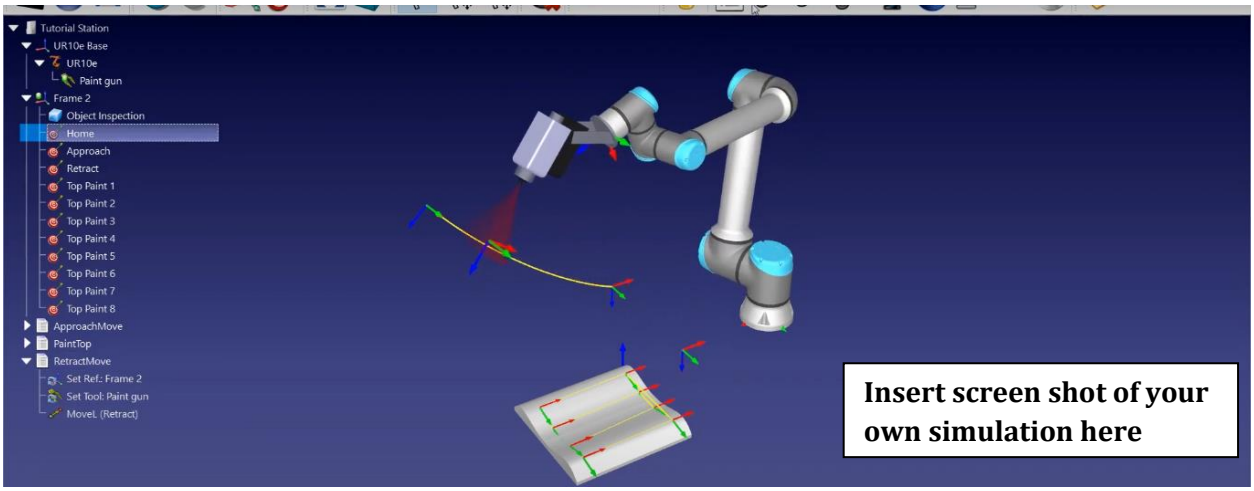
For Robot simulation take following problem statement.

Problem Statement: - Simulate Spherical robot for Pick and Place application.

(Reference video for Simulation: - https://www.youtube.com/watch?v=P0rEwpmAHq4&t=2449s)

Take Robot configuration as per given in table according to your branch.

Sr. No.	Branch	Robot	Object	Pallet / Table
1	AIDS	ABB - CRB 1300-7/1.4	Box - 12x7in	Table - 108x114x32in
2	COMP	ABB - CRB 1300-10/1.15	Bin Blue	Pallet - 1200x800mm
3	IT	ABB - IRB 140-6/0.8	Bin Pink	Table - 1400x800x800mm
4	ELECT	KUKA - KR 4 R600	Box - 12x7in	Pallet - 1625x1350mm
5	ENTC	KUKA - iisy 3 R1300	Bin Red	Table - 108x114x32in
6	MECH	KUKA - KR 5 arc	Box - 12x7in	Pallet - 1200x800mm
7	CIVIL	KUKA - KR 5 sixx R850	Box - 12x7in	Pallet - 1200x800mm



Export pdf simulation and Export Robot Programming (Take print and attached with this), Refer following video for exporting pdf simulation and generating programming.

https://www.youtube.com/watch?v=cPdMY4IO0cY

Conclusion:-

Simulating spherical robots using software like RoboDK is a crucial step in the design and development process. Each robot configuration has distinct characteristics and is best suited to specific applications, whether it's linear movement in Cartesian robots, cylindrical motion in cylindrical robots, or spherical coverage in spherical robots.

Simulation tools offer a safe and cost-effective way to evaluate robot performance, optimize movements, and prevent collisions before physical implementation. These tools are indispensable for researchers and engineers to visualize, analyze, and refine robotic systems for both educational and industrial applications.

EXPERIMENT NO. 4

Title:-

Demonstration of SCARA robot.

Objective: -

The objective of this study is to comprehensively analyze the design, motion characteristics, work envelope, and industrial applications of SCARA robots. By examining their structural components and motion capabilities, the study aims to illustrate how SCARA robots achieve high-speed and high-precision operations, especially in tasks requiring selective compliance in the horizontal plane. Through practical demonstrations, such as color sorting and weight-based sorting applications, this study seeks to highlight the potential of SCARA robots to improve efficiency, accuracy, and productivity in manufacturing and assembly processes.

Introduction:-

The **SCARA robot** (Selective Compliance Articulated Robot Arm) is a type of industrial robot that is designed for high-speed, high-precision tasks in manufacturing and assembly processes. It is characterized by its selective compliance, which allows flexibility in the horizontal plane while maintaining rigidity in the vertical direction. This unique feature makes SCARA robots ideal for tasks such as pick-and-place operations, assembly, and material handling.

The SCARA robot operates in a cylindrical work envelope, which gives it a compact design with excellent speed and precision. In this study, we will explore the structure, motion, work envelope, and key applications of SCARA robots.

---

1. Structure of SCARA Robots

- **Arm Configuration:**
  - The SCARA robot has a two-link arm structure, consisting of two rotational joints and a vertical prismatic joint.
  - The first two joints provide rotational motion in the horizontal plane, allowing the arm to extend, retract, and rotate.
  - The prismatic joint provides vertical movement, enabling the robot to move up and down.
- **Axes of Motion:**

- **Theta 1 (θ_1):** The first rotational joint allows the robot arm to rotate around the base, giving it the ability to reach different positions within the workspace.
- **Theta 2 (θ_2):** The second rotational joint enables further extension or retraction of the arm along the radial axis.
- **Z-axis:** The prismatic joint provides vertical movement, allowing the robot to reach different heights.

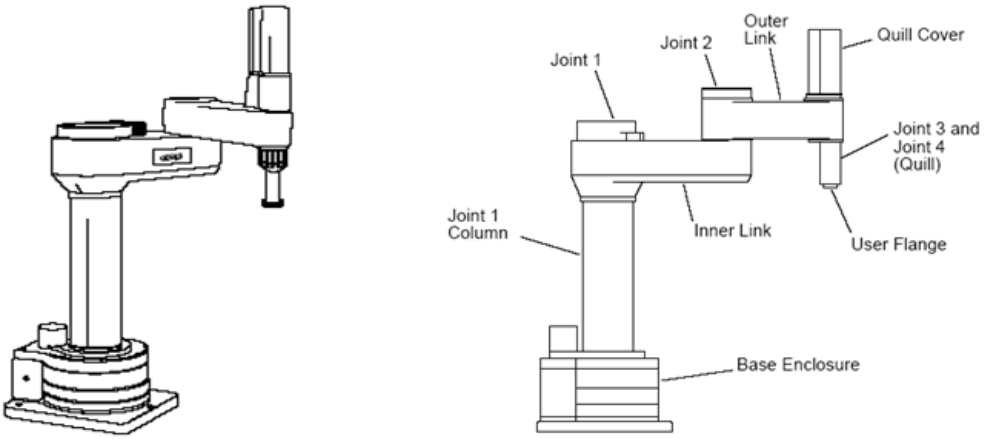


Figure 1: - Scara Robot Configuration.

2. Motion of SCARA Robots

- **Planar Motion:**
  - SCARA robots exhibit high-speed motion in the horizontal plane (XY plane), making them ideal for tasks that involve lateral movement. The flexibility in this plane allows the robot to move objects quickly between different locations.
- **Vertical Movement:**
  - The vertical prismatic joint provides controlled motion along the Z-axis, allowing the robot to lift and lower objects with precision.
- **Selective Compliance:**
  - SCARA robots are designed to have compliance in the horizontal plane, which means they can adapt to slight variations in object position, making them excellent for assembly tasks.
  - However, they maintain rigidity in the vertical direction, ensuring precise up-and-down movements.

3. Work Envelope of SCARA Robots

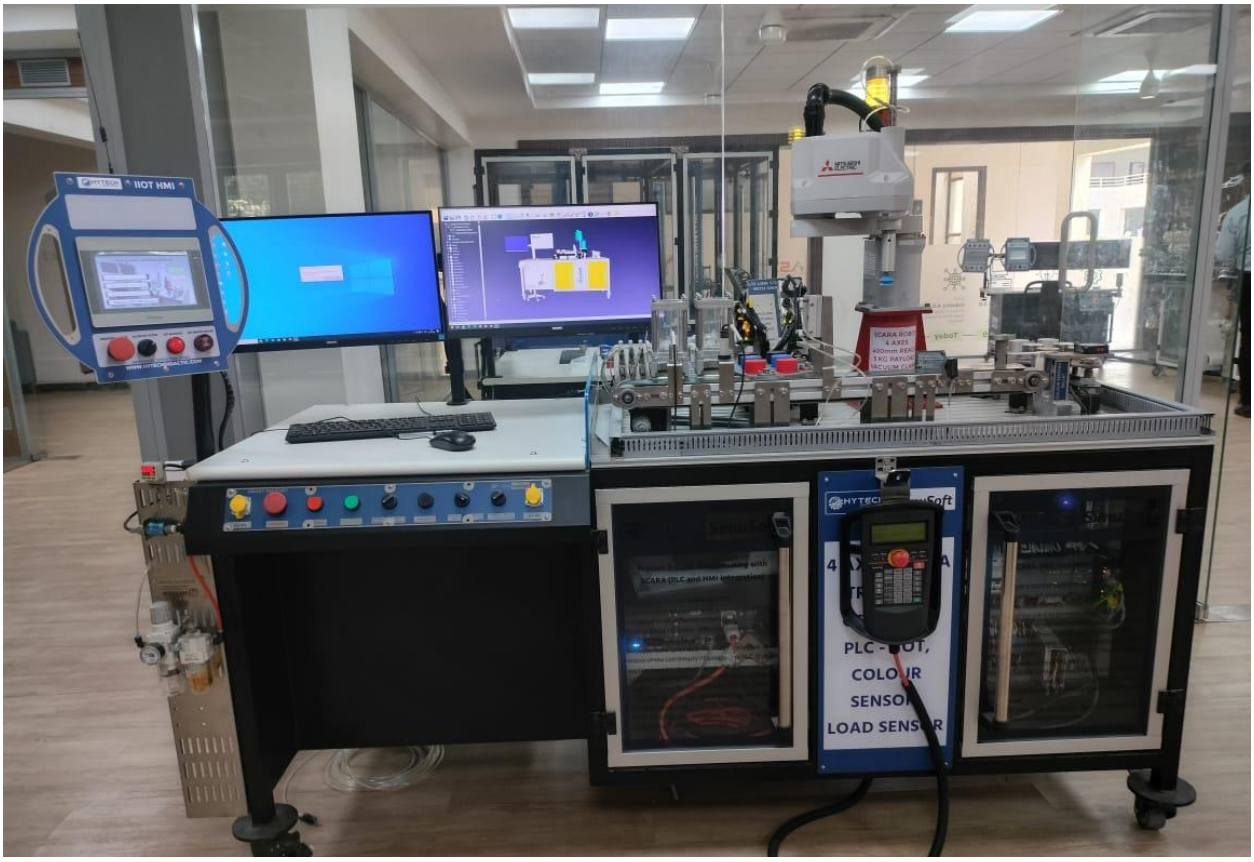
- **Cylindrical Workspace:**
  - The workspace of a SCARA robot is cylindrical, with a wide range of motion in the horizontal plane due to the rotational joints. The vertical prismatic joint allows for linear movement along the Z-axis.
  - The **radial reach** is defined by the combined length of the two rotational arms, while the **vertical reach** is limited by the range of the prismatic joint.
- **Compact Design:**
  - The compact cylindrical workspace allows the SCARA robot to be used in environments with limited space, such as small workstations or assembly lines.

4. Applications of SCARA Robots

SCARA robots are primarily used in industries where speed, precision, and repeatability are crucial. Some of the key applications include:

- **Pick-and-Place Operations:**
  - SCARA robots excel at high-speed pick-and-place tasks, where objects need to be transferred from one location to another with precision.
- **Assembly:**
  - These robots are ideal for assembling small components in electronics, automotive, and consumer goods manufacturing due to their accuracy and flexibility in the horizontal plane.
- **Material Handling:**
  - SCARA robots can quickly handle lightweight materials in production lines, transferring items between conveyor belts, pallets, and assembly stations.
- **Packaging:**
  - In packaging operations, SCARA robots are used to arrange, sort, and pack items in boxes or trays, ensuring consistency and speed.
- **Soldering and Welding:**
  - SCARA robots are sometimes used in precision soldering or light welding applications, especially in electronics manufacturing.

Demonstration on 4 Axes SCARA Robot Training Cell:-



1. Color Sorting Application

'PICK FROM CONVEYER - PLACE AT COLOR SENSOR

Coordinates

Position

Add | Edit | Delete | Robot: 1:RH-3CRH4018-D | Get Current Pos. | Pos. Jump

Name	X	Y	Z	A	B	C	L1	L2	FLG1	FLG2
PCOLORSEN...	296.229	253.926	21.554	0.000	0.000	-122.070	X	X	4	0
PCOLORSEN...	296.229	253.926	133.234	0.000	0.000	-122.054	X	X	4	0
PHOME	133.314	-63.701	144.562	0.000	0.000	-108.805	X	X	4	0
PICK	357.228	-10.969	7.559	0.000	0.000	-141.258	X	X	4	0
PICKSAFE	357.228	-10.969	134.715	0.000	0.000	-141.258	X	X	4	0

Program

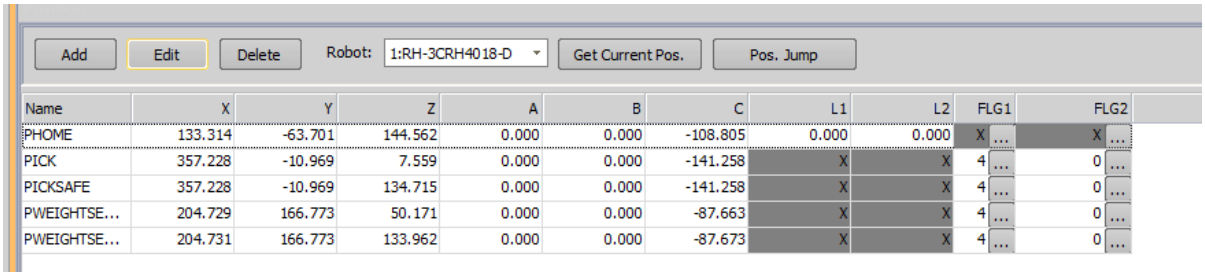
```
Ovrd 8
'Mov PHOME
Mov PICKSAFE
Ovrd 8
```

Mov PICK
Dly 1
M_Out(4)=1 'GRIPPER CLAMP
Dly 1
Mov PICKSAFE
Ovrd 8
Mov PCOLORSENSORSAFE 'SAFE LOCATION ABOVE SOLOR SENSOR
Ovrd 8
Mov PCOLORSENSOR
Dly 1
M_Out(4)=0 'GRIPPER DECLAMP
Dly 1
Mov PCOLORSENSORSAFE
Ovrd 8
'Mov PHOME
End

2. Weight Base Sorting Application

'PICK FROM CONVEYER - PLACE AT WEIGHING STATION

Coordinates



Name	X	Y	Z	A	B	C	L1	L2	FLG1	FLG2
PHOME	133.314	-63.701	144.562	0.000	0.000	-108.805	0.000	0.000	X	X
PICK	357.228	-10.969	7.559	0.000	0.000	-141.258	X	X	4	0
PICKSAFE	357.228	-10.969	134.715	0.000	0.000	-141.258	X	X	4	0
PWEIGHTSE...	204.729	166.773	50.171	0.000	0.000	-87.663	X	X	4	0
PWEIGHTSE...	204.731	166.773	133.962	0.000	0.000	-87.673	X	X	4	0

Program

Ovrd 8
'Mov PHOME
Mov PICKSAFE
Mov PICK
Dly 1
M_Out(4)=1 'GRIPPER CLAMP
Dly 1
Mov PICKSAFE

```
Mov PWEIGHTSENSOR_SAFE ' SAFE LOCATION ABOVE WEIGHT BLOCK
Mov PWEIGHTSENSOR
Dly 1
M_Out(4)=0 'GRIPPER DECLAMP
Dly 1
Mov PWEIGHTSENSOR_SAFE
'Mov PHOME
End
```

Conclusion:-

SCARA robots are versatile and efficient solutions for various high-speed, high-precision industrial tasks, including pick-and-place operations, assembly, and material handling. Their unique design allows them to achieve selective compliance, adapting well to horizontal movements while maintaining stability in the vertical plane. This selective compliance and compact cylindrical work envelope make SCARA robots particularly valuable in environments with space constraints or requirements for precise, repetitive actions. As demonstrated in training applications, such as color sorting and weight-based sorting, SCARA robots can streamline manufacturing processes by automating tasks with speed, accuracy, and reliability. Their continued development and integration in industries promise advancements in efficiency and productivity for complex assembly and manufacturing operations.

EXPERIMENT NO. 5

Title:-

Simulation & Demonstration of a simple robotic system using RoboDK software for material handling systems.

Objective: -

The goal of this study is to:

1. Examine the design and working principles of a robotic material handling system.
2. Understand how robots perform tasks like picking, placing, and transferring objects.

Introduction:-

Material handling systems are integral to manufacturing, warehousing, and logistics operations. They are used to move, store, and manage goods efficiently. A robotic system in material handling combines automation with precision, offering improved efficiency, reduced labor costs, and greater consistency. This study focuses on understanding the components, functionality, and benefits of a simple robotic system designed for material handling tasks.

Components of the Robotic Material Handling System

A basic robotic system for material handling consists of several key components:

1. Robotic Arm

- **Type**: SCARA, 6-axis industrial robots, or Cartesian robots.
- **Function**: Performs pick-and-place operations with precision.
- **Capabilities**: Flexible movement along multiple axes for reaching and manipulating objects.

2. End Effector

- **Type**: Gripper, vacuum suction, or magnetic tool.
- **Purpose**: Secures objects during handling.
- **Selection Criteria**:
  - Object shape, size, and weight.
  - Sensitivity of the object (fragile or robust).

3. Conveyor Belt

- **Purpose**: Transports objects to and from the robotic arm's pick-up and drop-off locations.
- **Integration**: Works with sensors to synchronize the robot's actions with object positions.

4. Sensors

- **Types**:
  - Proximity Sensors: Detect object presence and trigger robot actions.
  - Vision Systems (Optional): Identify object orientation and ensure accurate handling.
- Role: Enhance automation by providing real-time feedback to the controller.

5. Control System

- **Type**: PLC (Programmable Logic Controller) or dedicated robot controllers.
- **Role**:
  - Processes input from sensors.
  - Executes predefined programs to control robot movements and actions.

6. Workstation Setup

- Includes pick-and-place points, storage racks, pallets, or trays for organizing handled materials.

Principle of Operation

The robotic material handling system operates in a sequence of steps:

1. **Object Detection**:
   - Sensors or vision systems identify the presence of objects on the conveyor belt.
   - The control system sends a signal to initiate the robot's action.
2. **Pick Operation**:
   - The robot moves to the pick position based on pre-programmed coordinates.
   - The end effector (e.g., gripper or vacuum tool) secures the object.
3. **Transition Movement**:
   - The robot lifts the object and moves it through a predefined safe path to avoid collisions.
   - Intermediate points can be programmed for smooth transitions.
4. **Place Operation**:

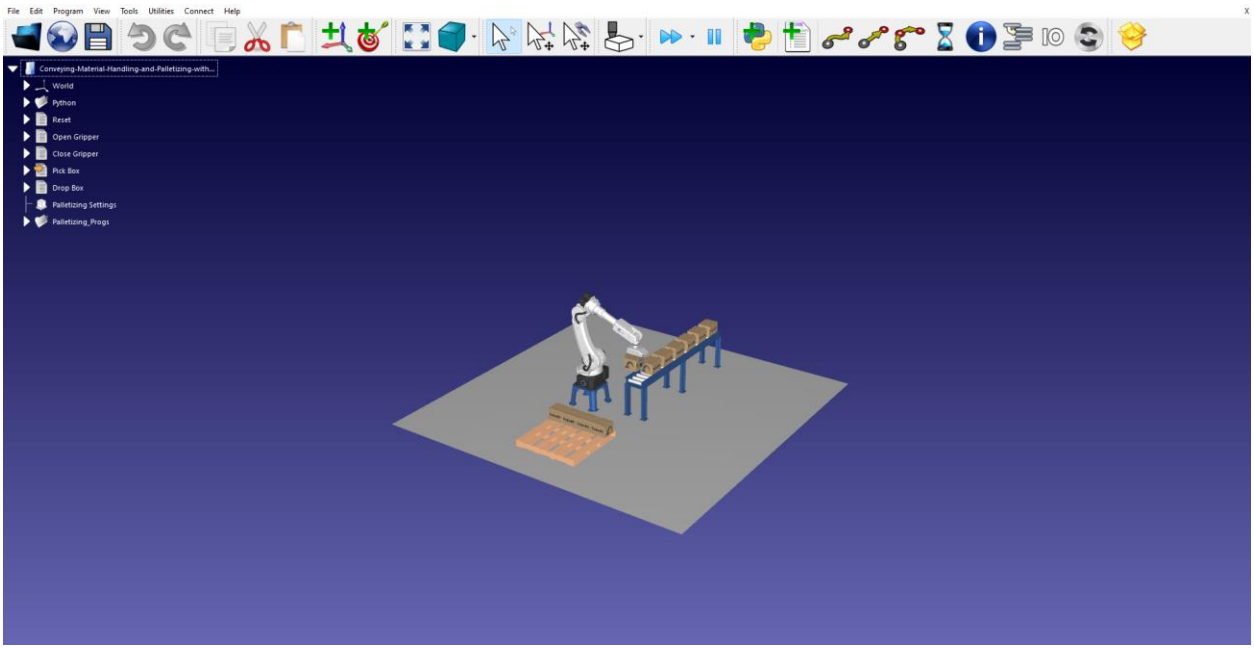
- The robot positions the object at the designated location (e.g., pallet, rack).
- The end effector releases the object securely.

5. **Return to Home Position**:
    - The robot resets to its home position, ready for the next cycle.

Programming the Robotic System

The robot is programmed to perform tasks using predefined commands and routines. Key aspects of programming include:

1. **Defining Coordinates**:
    - Precise locations for pick, place, and transition points are programmed.
2. **Motion Paths**:
    - The robot's movement is defined as either linear (straight-line motion) or joint (smooth, rotational paths).
3. **Gripper Control**:
    - Commands to activate or deactivate the end effector at the appropriate stages.
4. **Sensor Integration**:
    - Input from sensors triggers specific robot actions dynamically.





Applications of the System

1. Pick-and-Place Operations:

- The robot picks items from the conveyor and places them on pallets, trays, or assembly lines.

2. Sorting:

- Based on weight, size, or color, the robot sorts objects into designated bins or locations.

3. Assembly:

- Assists in assembling components by placing them in precise positions.

4. Packaging:

- Arranges items in boxes or cartons in an organized manner.

5. Loading and Unloading:

- Transfers goods between conveyors, racks, and storage areas.

Advantages of Robotic Material Handling

1. **Increased Efficiency**:
    - Robots operate at high speeds with consistent accuracy.
2. **Reduced Labor Costs**:
    - Automates repetitive tasks, reducing dependency on manual labor.
3. **Improved Safety**:
    - Minimizes human involvement in hazardous environments.
4. **Flexibility**:
    - Easily adaptable to different tasks by reprogramming.
5. **Precision**:
    - Capable of handling delicate and lightweight objects with minimal errors.

Conclusion:-

The study of a simple robotic system for material handling highlights its ability to enhance efficiency, precision, and safety in industrial operations. By automating repetitive and labor-intensive tasks, robotic systems contribute to improved productivity and reduced operational costs. With advancements in robotics and artificial intelligence, these systems are becoming more accessible and capable, paving the way for widespread adoption in diverse industries.

EXPERIMENT NO. 6

Title:-

Case study on selection & simulation of the appropriate robot system for given manufacturing and non- manufacturing applications.

Objective: -

1. To understand the process of selecting an appropriate robot system for specific tasks in manufacturing and non-manufacturing applications.

Introduction:-

The selection of a robotic system for any application requires careful consideration of task requirements, operational constraints, and system capabilities. By analyzing the specifics of both manufacturing and non-manufacturing tasks, this case study demonstrates how to select and simulate an appropriate robot system using defined criteria and simulation tools like RoboDK or other robotic software platforms.

Steps in Robot System Selection

1. Define Application Requirements

Identify the key parameters of the task to be automated:

- **Task Type**: Material handling, assembly, inspection, sorting, etc.
- **Payload**: Maximum weight the robot must handle.
- **Precision**: Required accuracy for the operation.
- **Speed**: Desired cycle time.
- **Environment**: Workspace layout, size constraints, and operating conditions (cleanroom, high temperatures, etc.).

2. Identify Suitable Robot Types

Based on the requirements, select the appropriate type of robot:

- **Articulated Robots**: Versatile, multiple DOF (Degrees of Freedom) for complex tasks like welding or assembly.
- **SCARA Robots**: High-speed, high-precision operations for assembly and pick-and-place tasks.

- **Delta Robots**: Lightweight, high-speed tasks like packaging and sorting.
- **Cartesian Robots**: Linear motion tasks such as CNC machining and 3D printing.
- **Collaborative Robots (Cobots)**: Safe for working alongside humans, ideal for mixed environments.

3. Analyze Feasibility

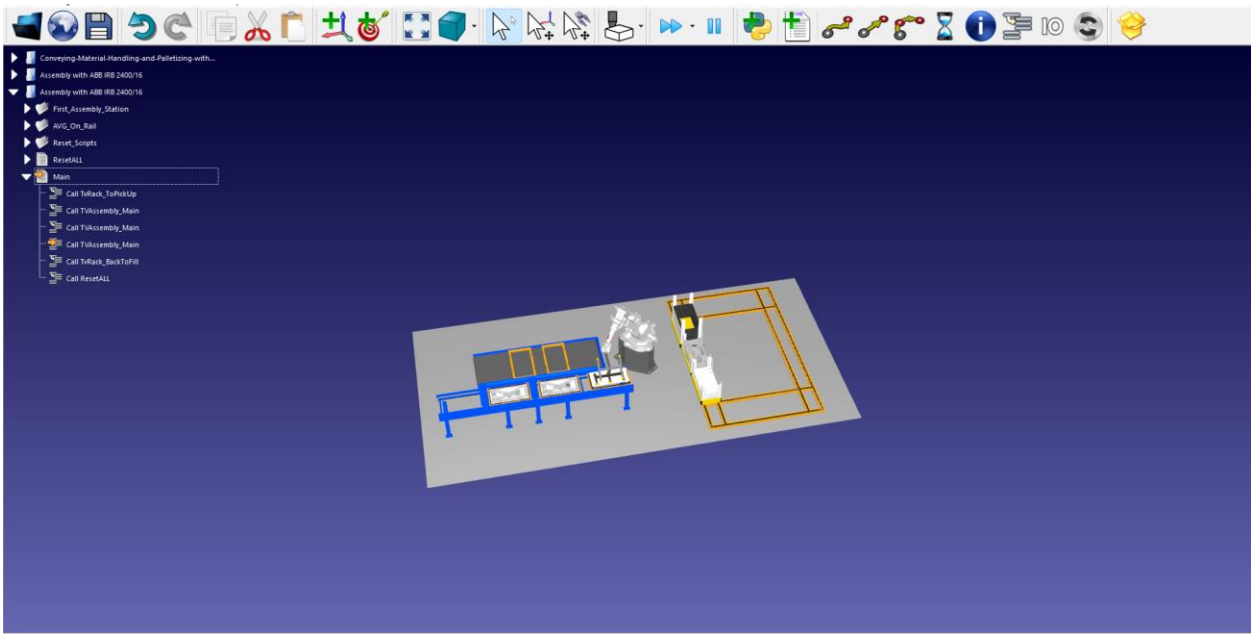
Evaluate the selected robot's compatibility with:

- **Workspace Constraints**: Check if the robot fits within the operational area.
- **Payload Capacity**: Ensure the robot can handle the maximum load.
- **Reach**: Validate if the robot can access all necessary points.
- **Cost-Effectiveness**: Compare investment cost against projected ROI.

Case Scenarios

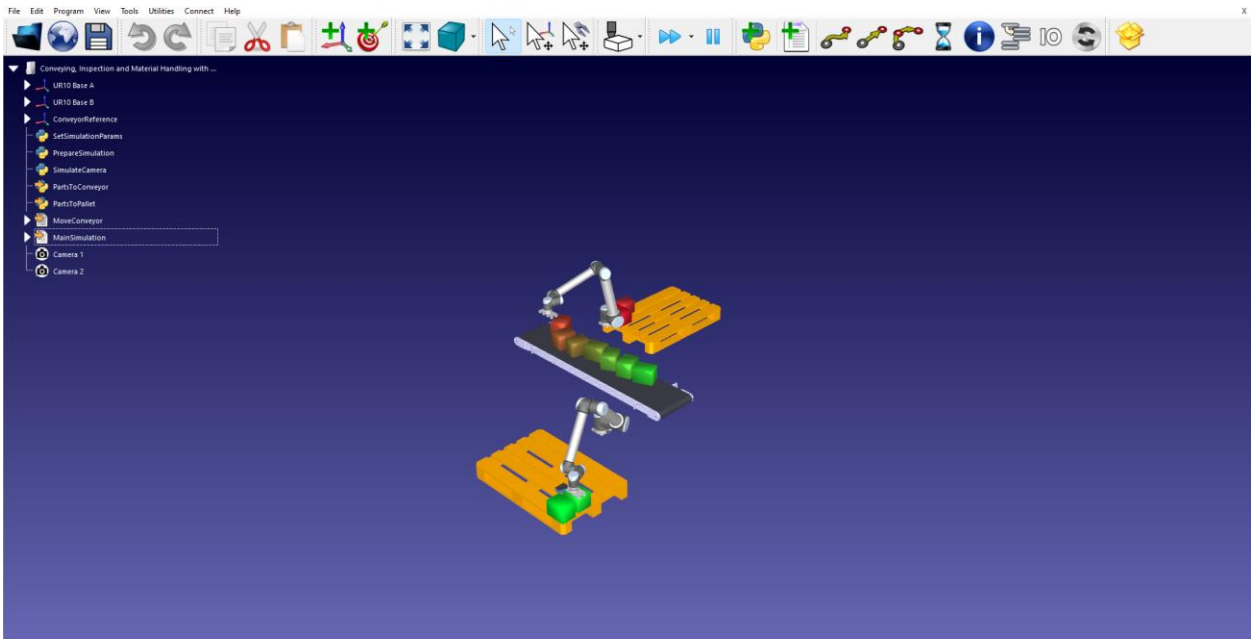
Manufacturing Application: Assembly of Electronic Components

1. **Task Description**:
   Assemble small electronic components on a circuit board.
2. **Requirements**:
   - **Precision**: ±0.1 mm.
   - **Speed**: High throughput (200 components/hour).
   - **Payload**: ≤ 5 kg.
   - **Environment**: Cleanroom for electronic manufacturing.
3. **Robot Selection**:
   - **Type**: SCARA Robot.
   - **Reason**: High-speed planar motion with high precision. Compact size fits the cleanroom environment.
4. **Simulation**:
   - **Software:** RoboDK.
   - Simulated task flow:
     a. Robot picks components from a feeder using a vacuum gripper.
     b. Places them on the circuit board at precise coordinates.
     c. Repeats for the entire assembly.
5. **Output**:
   Simulation validated robot precision, ensured no collisions, and met the required cycle time.



Non-Manufacturing Application: Automated Sorting in a Warehouse

1. **Task Description**:
   Sort and place parcels based on their weight and size into designated bins.
2. **Requirements**:
   - **Payload**: 20 kg.
   - **Speed**: Moderate throughput (50 parcels/hour).
   - **Precision**: ±1 mm.
   - **Environment**: Open warehouse with variable lighting.
3. **Robot Selection**:
   - **Type**: Collaborative Robot (Cobots).
   - **Reason**: Flexible, safe for human interaction, and can handle parcels without extensive guarding.
4. **Simulation**:
   - Software: RoboDK or equivalent.
   - Simulated task flow:
     a. Sensors identify parcel dimensions and weight.
     b. Cobots grip the parcel and place it in the corresponding bin.
     c. System adapts to varied parcel sizes dynamically.
5. **Output**:
   Simulation confirmed the cobot's ability to sort parcels within the required cycle time. Safety checks ensured compliance with collaborative workspace standards.



Selection Criteria for Robot Systems

Criteria	Manufacturing	Non-Manufacturing
Task Complexity	High (precision assembly)	Moderate (sorting and handling)
Payload	Low (light components)	Moderate (parcels up to 20 kg)
Required Speed	High	Moderate
Safety Requirements	Guarded environment	Collaborative workspace
Cost Consideration	Medium to high	Medium

Conclusion:-

The selection and simulation of robotic systems are crucial for successful automation in both manufacturing and non-manufacturing tasks. This case study highlights the importance of identifying task requirements, evaluating robot types, and ensuring feasibility through simulation. SCARA robots excel in precision assembly for manufacturing, while collaborative robots (cobots) are ideal for adaptive and safe operations in non-manufacturing scenarios like warehouse sorting. Advanced simulation tools validate performance and optimize operations, enabling organizations to enhance efficiency, accuracy, and productivity.